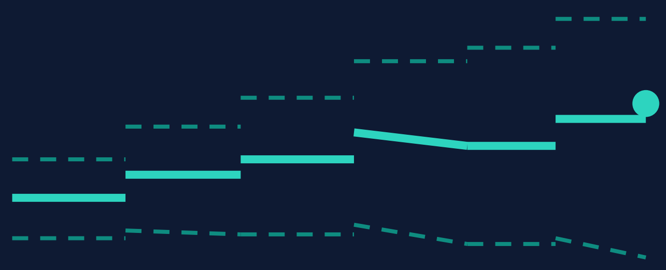
SYBILION HACKATHON 2026 · FORECASTING AI TRACK

Infineon Cost & Treasury Agent

Forecast → Hedge → Stock Safety

A decision agent on the Sybilion probabilistic-forecasting API that scopes a chip maker's USD-priced cost exposures — FX, wholesale power and commodities — forecasts each, keeps only the high-confidence signals, and turns them into a hedge or a safety-stock buffer, every recommendation walk-forward backtested on real data.

6-MONTH FORECAST · P10 / P50 / P90



TEAM

43 Food

Fridolin Sitter · Elliot Sezalory

01 · THE PROBLEM

Exposed on Three Fronts at Once

A European power-semiconductor maker like Infineon buys in USD but reports in EUR — and its margin is hostage to inputs it doesn't control. A single point forecast can't answer the treasury question:

THE TREASURY QUESTION

Which exposure is worth acting on, with **what conviction**, taking **what action** — and would acting on it **historically have helped?**

Why it matters: the interesting problem is **multi-exposure** — the same forecast bands must drive three different decisions, and the agent has to decide which and when, including when the honest answer is *“bands too wide, don’t act.”*

WHERE THE MARGIN LEAKS

FX

USD-denominated supplier payments vs a EUR book

Wholesale power

Energy-intensive fabs exposed to day-ahead prices

Commodities

Copper, gallium and gas feed the cost basis

Four Layers, From Prompt to Proven Action

Layer	What it does	Output
1 SCOPE Qwen2.5-7B · calendar	Featherless Qwen + a live economic calendar pick the exposures worth forecasting.	**D1** Exposures to forecast
2 PREPARE FRED · Pink Sheet · APG	Build monthly histories and normalize each series into the Sybilion API's shape.	Sybilion-ready series
3 FORECAST Sybilion API	Returns a 6-month probabilistic forecast, then we score each signal's confidence.	**D2** STRONG / WEAK signal
4 STRATEGY Claude Sonnet · rules	On STRONG signals only: map to an action and walk-forward backtest it vs the realized series.	FX forward Power hedge Safety-stock buffer — END GOAL

ONE SCORE GATES EVERYTHING `conf = 100·tanh(move / band)·(1 − mape)` → STRONG only when the move beats the band

Designing for Uncertainty, Not Around It

ONE CONFIDENCE SCORE, *three decisions*

A signal is STRONG only when the predicted move is large relative to the forecast's own uncertainty. The same band width also sizes the safety-stock buffer.

```
band   = (p90 − p10) / p50
move   = |p50 − now| / now
conviction = move / band
conf   = 100·tanh(conviction)
          ·(1 − min(mape,50)/50)
```

Wide bands → more buffer weeks. **Directional bias → pre-buy vs run lean.**

Forecast as a filter

For power & commodities we don't trade the forecast — we use Sybilion's direction to filter a Donchian breakout, then measure filtered vs unfiltered.

Walk-forward, never in-sample

Every backtest truncates history and scores against the realized next value, so accuracy is genuinely out-of-sample.

Mock-first architecture

Mock and live clients share one shape: the app runs offline with zero keys and swaps to live Sybilion with a single flag.

Shock scenarios

boj_hike · ecb_cut · china_deval re-bias the forecasts mid-run, so signals and actions visibly flip on stage.

05 · RESULTS — FX DIRECTION SKILL

We Report the Mock-to-Live Gap Honestly

83%

Mock forecasts

6 signals · 2026-05 · STRONG-only 2 / 2

40%

Live Sybilion

30 signals · 2026-01 → 2026-05

WHAT THE NUMBER MEANS

The polished demo (83%) and live reality (40%) are both shown — no cherry-picking. STRONG signals still beat WEAK in the mock set, and the confidence gate keeps the agent on the sidelines when the bands are too wide to justify a trade. Currency direction at a 1-month horizon is simply hard; the value is the discipline around the forecast, not a magic hit-rate.

THE HONEST READ

Monthly univariate forecasts are a **thin directional edge**. Across the live FX set we hit **40%** — roughly coin-flip at the monthly horizon for currencies.

Why the gap? We submit with `filters.limit = 0` — pure univariate, no external driver signals. That leaves Sybilion's most decision-relevant output on the table.

Artifacts: monthly_backtest_live.json · pairs28_h6_live.json

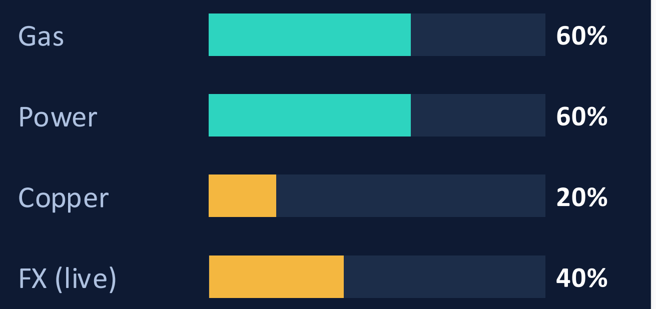
On Power & Commodities, the Filter Clearly Helps

Asset	Forecast	Baseline Donchian	+ Sybilion filter	Dir.
Copper · 132m	DOWN	10 trades · +0.9%	8 trades · −0.0% · ½ drawdown	20%
TTF gas · 132m	UP	11 trades · −2.5%	4 trades · −0.2% · cut losers	60%
AT power · 137m	UP	10 trades · −0.4%	3 trades · +0.3% · flipped +	60%

The win: the direction filter cuts trade count and drawdown on every asset, and on Austrian power it flips a losing strategy positive (−0.4% → +0.3%). The forecast earns its place as a ***filter*** — not as something to trade directly.

The honest caveat: none of these beat naive buy-and-hold over the long windows (copper +154%, power +254%). But “buy and hold a cost input forever” isn't a treasury action — the right comparison is filtered vs unfiltered, where the forecast clearly helps.

NEXT-MONTH DIRECTION HIT



~coin-flip on copper & FX; real skill on gas & power.

Band-width → safety stock

For materials we don't trade at all — the forecast's uncertainty sizes the buffer directly: wider bands buy more weeks of safety stock, a directional bias means pre-buy vs run lean.

The Clearest Path Past 40%

PRIORITY

Turn on external drivers

Submit with filters.limit > 0 and surface Sybilion's driver-importance per horizon — copper ← China export controls, power ← gas + carbon, FX ← rate differentials. Directly serves “visible reasoning,” and the likeliest path to beat 40%.

Cross-exposure synthesis

A USD spike raises both the FX-hedge need and material cost. Let the agent reason across exposures at once instead of one at a time.

Confidence-scaled sizing

Make hedge ratio and buffer weeks proportional to the score, plus a calibration report: are 70%-confidence signals right 70% of the time?

Longer live track record

Extend beyond 5 months and put transaction costs into the P&L for an honest, deployable read.